

Information Technologies for Industrial Engineers

Git

What is Git?

- Git is a distributed **version control system** that allows multiple developers to work on a project simultaneously.
- It tracks changes to files and allows developers to collaborate on code, manage different versions of a project, and revert to previous versions if needed.
- Git is widely used in software development and is the foundation for platforms like `GitHub`, `GitLab`, and `Bitbucket`.

Get started with Github

- Create a free account on [GitHub](#)
- Install Git on your computer
 - [Downloads](#)
 - `choco install git` (Windows)
 - `brew install git` (macOS)
- Configure Git with your username and email

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

Useful git commands

- `git init` - Initialize a new Git repository
- `git clone <repository-url>` - Clone an existing repository
- `git add <file>` - Stage changes for commit
- `git commit -m "commit message"` - Commit staged changes with a message
- `git push` - Push committed changes to a remote repository
- `git pull` - Fetch and merge changes from a remote repository
- `git status` - Show the status of the working directory and staging area

Exercise 1

- Create a new repository on GitHub
- Clone the repository to your local machine
- Create a new file, make some changes, and commit the changes
- Push the changes back to GitHub

.gitignore

- A `.gitignore` file specifies intentionally untracked files that Git should ignore. Files already tracked by Git are not affected.
- Example of `.gitignore` in Node.js project ([Link](#))

Exercise 2

- Create a `.gitignore` file in your repository
- Add the following lines to ignore

```
node_modules/
```

- Run `pnpm install express` to create a `node_modules` folder
- Check the status of your repository using `git status`.

Git Remote

- A remote is a version of your project that is hosted on the internet or network somewhere.
- The most common remote is `origin`, which is the default name for a remote repository when you clone a repository.

Git Remote Commands

- `git remote -v` - List remotes
- `git remote add origin <repository-url>` - Add a remote
- `git remote remove origin` - Remove a remote
- `git remote set-url origin <new-url>` - Update a remote URL

Clone Backend Template

- For your future backend projects, you can clone the backend template repository to get started quickly.
- Clone the backend template repository

```
git clone https://github.com/it-for-ie-69/express-template <folder-name>
```

- Replace the template repository remote with your new repository, then push your work

```
git remote rename origin upstream  
git remote add origin <your-new-repository-url>  
git branch -M main  
git push -u origin main
```